

Lab: Cryptanalysis



**Skills
Network**

Estimated time: 3 minutes

Introduction

Cryptanalysis is the process of decoding encrypted information without encryption key.

In this lab, you'll explore the Caesar cipher, a simple encryption method that shifts each letter in the plaintext by a fixed number of positions in the alphabet. Using the cryptii tool, you'll decrypt messages encoded with the Caesar cipher, practicing techniques such as brute-forcing and manual decryption. You'll also identify how the Caesar cipher works, learn how to break it using known shifts, and see its vulnerabilities. Through hands-on activities, you'll gain experience analyzing and decoding encrypted messages.

Lab objectives

By the end of this lab, you will be able to:

- Identify the working principle of the Caesar cipher and how it encrypts and decrypts messages
- Apply the decryption process using a known shift value to retrieve the original plaintext from the ciphertext
- Use brute force and trial and error to identify the correct shift value when it is unknown
- Recognize the weaknesses of the Caesar cipher as a form of encryption and identify its vulnerabilities to simple cryptanalysis techniques
- Develop practical skills in performing cryptanalysis and learn how to identify patterns in ciphertext for decryption

This lab provides a foundation for exploring advanced cryptographic techniques while developing critical thinking and analytical skills needed for cybersecurity.

What is Caesar cipher?

The **Caesar cipher** is one of the oldest and simplest encryption techniques, named after Julius Caesar. It was believed that Julius Caesar used this Caesar cipher to protect his messages. It is a type of **substitution cipher**, where each letter in the plaintext is shifted by a certain number of positions in the alphabet.

For example, with a shift of 3:

- **A** becomes **D**
- **B** becomes **E**
- **C** becomes **F**, and so on.

To decrypt the message, shift the letters back by the same number. While the Caesar cipher was once useful for secret communications, it is now considered very insecure because it can be easily cracked by trying all possible shifts, such as brute force. Despite its simplicity, the Caesar cipher introduces the basic idea of encryption, which has led to the development of more advanced cryptographic methods used today.

Step-by-step guide for cryptanalysis using cryptii

Let's look at the step-wise guidance for cryptanalysis using cryptii.

Step 1: Open the cryptii tool

Go to [cryptii's website](#) to open the cryptii tool.

Step 2: Choose the cipher type

1. To choose ENCODE, select ENCODE on the cryptii homepage, and select encoding techniques.



Play the CODEBREAKERS game. The Enigma level is now unlocked. Find what is hidden.

VIEW
Text
The quick brown fox jumps over the lazy dog.

ENCODE

DECODE

Enigma machine

MODEL
Enigma M3
REFLECTOR
UKW B

ROTOR 1

VI

POSITION

- 1 A +

RING

- 1 A +

ROTOR 2

I

POSITION

- 17 Q +

RING

- 1 A +

ROTOR 3

III

POSITION

- 12 L +

RING

- 1 A +

PLUGBOARD

bq cr di ej kw mt os px uz gh

FOREIGN CHARS

Include Ignore

→ Encoded 41 chars

VIEW
Text
xasvj vvvgs jjtdp bknlj ruz

Modular conversion, encoding and encryption online

Web app offering modular conversion, encoding and encryption online. Translations are done in the browser without any server

2. You will see a list of cipher options. Select Caesar cipher from it.

cryptii.com

cryptii

Play the CODEBREAKERS game. The Enigma level is now unlocked. Find what is hidden.

VIEW

Text ▾

The quick brown fox jumps over the lazy dog.

Modular conversion, encoding and encryption online

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Library

Transform

- Replace
- Reverse
- Case transform
- Numeral system
- Bitwise operation

Alphabets

- Morse code
- Spelling alphabet

Ciphers

- Enigma machine
- Caesar cipher
- Affine cipher
- ROT13
- A1Z26
- Vigenère cipher
- Bacon cipher
- Alphabetical substitution
- Rail fence cipher

Polybius square ciphers

- Polybius square
- ADFGX cipher
- Bifid cipher
- Nihilist cipher
- Tap code
- Trifid cipher

Encoding

- Base32
- Base64
- Ascii85
- Baudot code
- Unicode code points
- URL encoding
- Punycode
- Bootstring
- Integer

Modern cryptography

- Block Cipher
- RC4
- Hash function
- HMAC

Step 3: Enter the encrypted text

1. In the left side of the input box, paste or type the **ciphertext** (the encrypted message) that you want to analyze.

For example: "Wklv lv d whvw phvvdjh."



Play the CODEBREAKERS game. The Enigma level is now unlocked. Find what is hidden.

VIEW
Text
Wklv lv d whvw phvvdjh

ENCODE DECODE
Caesar cipher
SHIFT
7 a→h
ALPHABET
abcdefghijklmnopqrstuvwxyz
CASE STRATEGY
Maintain case
FOREIGN CHARS
Include Ignore
→ Encoded 22 chars

VIEW
Text
Drsc sc k docd wocckqo

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Base32 Morse code with emojis Base32 to Hex Text to decimal Hex to ascii85

Step 4: Decrypt the cipher

- **Caesar cipher:** If you are using Caesar cipher, you can perform the decryption by selecting the **Shift** value (if you know the key).

- **Manual decryption:** You can enter the shift value directly to decrypt the message. Try various shifts (usually between 1 and 25) to see if one produces a readable message. **As you can see, with a shift value of 23, a readable plaintext is visible in the right text box.**

← → ↺ cryptii.com

 Play the CODEBREAKERS game. The Enigma level is now unlocked. Find what is hidden.

VIEW

+

Text ▾

Wklv lv d whvw phvvdjh

ENCODE DECODE

+

Caesar cipher ▾

SHIFT

— 23 a→x +

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY

Maintain case ▾

FOREIGN CHARS

Include Ignore

→ Encoded 22 chars

VIEW

+

Text ▾

This is a test message

Modular conversion, encoding and encryption online

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Base32 Morse code with emojis Base32 to Hex Text to decimal Hex to ascii85

Exercises

Exercise 1:

Ciphertext: Uifsf jt b tfdsfu dpef!

Task: Decrypt the ciphertext using the Caesar cipher.

Plain Text: There is a secret code!

Exercise 2:

Ciphertext: Byffi Zleyhxm

Task: Decrypt the ciphertext using the Caesar cipher.

Plain Text: Hello Friends

Exercise 3:

Ciphertext: Gojs Hvs Sbjwfcbasbh

Task: Decrypt the ciphertext using the Caesar cipher.

Plain Text: Save The Environment

Summary

In this lab, you've explored the **Caesar cipher**, a classical substitution cipher that shifts letters in the alphabet by a fixed number. Using the **cryptii tool**, you've demonstrated how to encrypt and decrypt messages using the Caesar cipher. The exercises helped you to understand how the cipher works by shifting each letter of the ciphertext by a specific number of positions. You've also focused on applying a known shift value to decode the ciphertext and retrieve the original plaintext.

The decryption process involves systematically trying different shift values or using brute force when the shift is unknown, allowing us to identify patterns and uncover the hidden message. The exercises provided in this lab showcase the practical experience in using cryptanalysis techniques for breaking Caesar ciphers.

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